Channel- and Task-Aware Object-Level Communication for Bandwidth-Constrained V2V Cooperative Perception

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Abstract—A vehicle-to-vehicle cooperative perception system must decide, under a mean per-frame communication budget, what its collaborator should transmit. Transmitting a feature map on every frame costs 5.0× the budget; transmitting object-level detections on every frame spends communication on frames whose detection would not change. We present CA-TOSG, a receiver-driven per-frame selector that chooses the transmitted representation from ego-local perception cues and an estimated channel state, signals the choice with a two-bit request, and charges every message through a measured LDPC/QAM chain.

At matched realised payload, CA-TOSG exceeds a random selector, an ego-cue threshold and an SNR threshold by +0.00798, +0.00575 and +0.00080 in scene-equal F_1 on Test, with every pairwise 95% scene-level bootstrap lower bound above zero; on Culver-City it attains the highest point estimate and the interval against the SNR threshold includes zero. Against always transmitting object-level detections it reduces cooperative-perception payload by 40.68% and 41.30% at scene-equal F_1 differences of -0.00036 and -0.00133. Holding the number of lost codewords fixed and varying which codewords are lost changes the detection outcome on 28.48% of matched pairs, so transport loss affects perception through its position and not only its quantity. The contributions are the receiver-driven pre-request formulation, a per-frame transport-accounting protocol, and these three measured results.

Index Terms—Vehicle-to-vehicle communication, cooperative perception, semantic communication, bandwidth adaptation, 3D object detection.

I. Introduction

Connected and automated vehicles exchange perception information over vehicle-to-vehicle (V2V) links to recover objects a single vehicle cannot see—at intersections, in dense traffic, behind occlusions and at long range [1], [2]. What the collaborator should transmit is not settled. Late fusion sends object-level detections [1]: compact, but discarding the intermediate representation. Intermediate fusion sends feature maps [2], [3]: richer, at a payload that recent work reduces by spatial selection, masking or compression [3] and that semantic communication optimises for task performance [4], [5].

Two constraints bind that choice. First, a V2V link carries a mean payload budget: under the budget used here, transmitting a feature map on every frame costs 5.0× what is available. Second, the link is time-varying [6], [7], so a message that would help on a clean channel can arrive damaged, and the payload spent on it is wasted. The useful representation therefore depends on both the perception state of the frame and the state of the channel, and it is a per-frame quantity.

Existing methods select content within a fixed representation and are evaluated against declared payload budgets. Two gaps follow. A method that varies the representation itself needs inputs available before the message is requested, or the decision depends on the message it is deciding about. And a payload claim needs the transmitted bits charged through the transport chain they traverse, or the comparison rests on a constant that no measurement produced.

This paper addresses both. CA-TOSG forms a 23-dimensional input from 21 ego-local perception and collaborator-availability cues and 2 channel cues, predicts a communication action with a random forest, and encodes the action as a two-bit request; the collaborator returns the requested representation and the ego fuses it. Every payload is charged per frame through a measured LDPC/QAM chain. The candidate action set is $\mathcal{A} = \{E, L, F\}$—ego-only, object-level and feature-level—and the policy frozen on the development split selects E or L.

Our contributions are three.

- A receiver-driven per-frame selection formulation in which every selector input is ego-local and exists before the request is issued, so no input depends on the message being requested. The objective is the Lagrangian relaxation of a budget-constrained perception–communication problem (Section III-G), with the budget a constraint on the mean payload.
- A per-frame transport-accounting protocol: each message is packetised, coded and modulated, and charged in channel uses, so payload comparisons are measured rather than declared (Section III-E).
- Three measured results under one detector, one field of view and one ground truth: at matched realised payload CA-TOSG exceeds random, ego-cue and SNR-threshold selection on Test (Section VI-C); it reduces cooperative-perception payload against always transmitting object-level detections (Section VI-C); and loss position at fixed loss amount changes the detection outcome (Section VI-E).

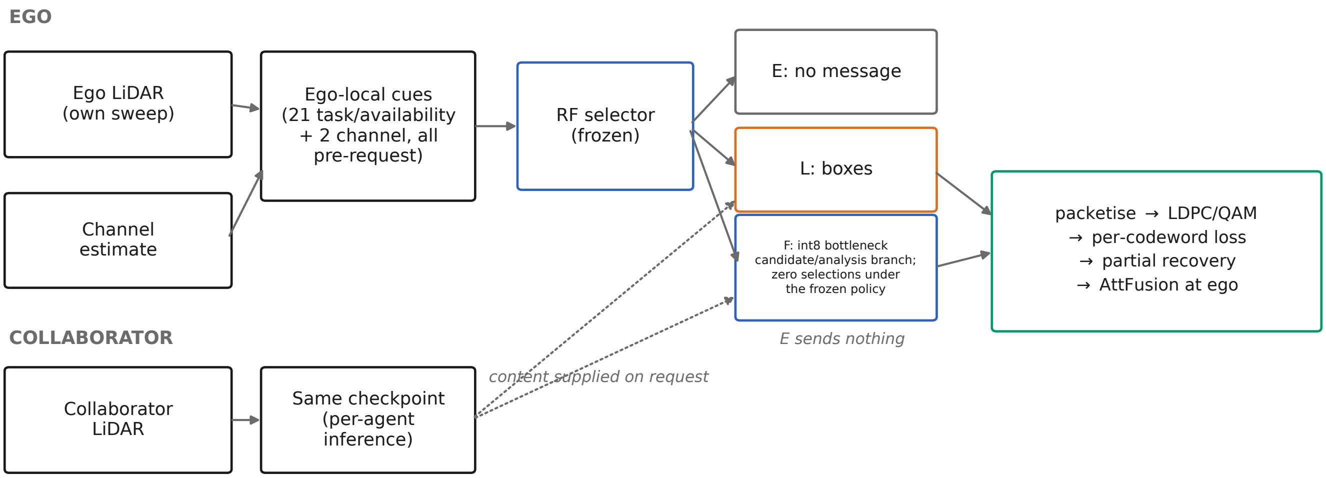

Fig. 1. Data flow as implemented. The cue vector and the channel estimate are ego-side and available before the request; the collaborator supplies message content only when asked. E issues no message and therefore never enters the transport chain.

II. Related Work

A. V2V Cooperative Perception

V2V cooperative perception exploits information from surrounding connected vehicles to alleviate occlusion, long-range sparsity and limited field of view [1]. Methods are categorised by the stage at which information is shared. Early fusion shares raw point clouds before feature extraction [1]: the most complete information, but a very large payload requiring accurate spatio-temporal alignment. Late fusion shares object-level predictions—boxes, classes, confidences [1]. The payload is small and suits low-rate links, but intermediate semantic and spatial information is discarded: an object missed by the sender cannot be recovered by the receiver. Intermediate fusion shares learned bird's-eye-view (BEV) features between backbone and detection head [2], [3], [8], [9], balancing the two extremes at a much larger message size. Benchmarks such as OPV2V and OpenCOOD [1] provide common datasets and protocols for comparing these strategies.

Relation to CA-TOSG. These works fix the representation and study fusion within it. CA-TOSG controls whether object-level cooperation is requested on each frame. Feature-level transmission is retained in the unified evaluation framework as a fixed-action upper bound and transport-analysis branch, but is not activated by the frozen deployed policy.

B. Communication-Efficient Cooperative Perception

To reduce the communication burden of intermediate fusion, many methods transmit only task-relevant features, using confidence maps, attention weights or importance masks to select informative spatial regions [3], [4], [10], [11]. Where2comm [3] is representative: it estimates where communication is needed and selectively transmits spatial feature regions. Feature compression is a second direction, encoding the map into a lower-dimensional representation before transmission [4], [12]. ML-Cooper [13] adapts the proportion of raw-, feature- and object-level data to the network state with a soft actor–critic policy, and RBH [14] chooses hybrid fusion by spatial overlap.

Relation to CA-TOSG. Where2comm selects which spatial positions to send inside the feature level; importance-map and JSCC methods optimise how the feature map is encoded and carried over the channel. ML-Cooper is the closest in spirit, in that it mixes levels, but it does so as a heavyweight learned resource-sharing policy with a different action mechanism. CA-TOSG acts one level above the question those methods answer: it controls whether object-level cooperation is requested on each frame. F is retained only as the upper-bound and transport-analysis branch described above.

C. Semantic and Channel-Aware Communication for Vehicular Perception

Semantic communication optimises the transmitted representation for task performance rather than for reconstruction fidelity or bit error rate [4], [5], [15]–[17]. Joint source–channel coding trains encoder and decoder with the channel in the loop and can degrade more gracefully than separated source and channel coding, whereas digital QAM+LDPC transport shows threshold behaviour [6]: poor below an SNR and rapidly reliable above it. That dependence matters most for vehicular links, whose SNR varies with multipath, blockage and fading [7]. Some work makes the transmission channel-adaptive at a fixed granularity: SmartCooper [12] adjusts the compression ratio from the estimated channel state, and AccBEV [18] conditions a feature-repair module on the SNR. CoDS [19] pairs a task-oriented semantic codec with a semantic analog-to-digital converter so that learned features traverse a standard digital V2X bitstream, and discards features corrupted by decoding failures at the receiver.

Relation to CA-TOSG. All of these adapt the encoding, the repair or the rate within feature-level transmission. CA-TOSG acts before that: it controls whether object-level cooperation is requested on each frame. F is retained only as the upper-bound and transport-analysis branch described above. The two lines of work are therefore complementary rather than competing: a digital semantic codec would occupy the feature branch this paper measures as an upper bound.

III. System Model and Problem Formulation

A. Cooperative Perception Setting

We consider a single ego–collaborator V2V link. At frame t the ego observes the scene with its own LiDAR and extracts a local BEV feature $F_{\text{ego},t}$ with a PointPillars backbone [20]; the collaborator observes from its own viewpoint and extracts $F_{j,t}$ with the same checkpoint. The ego fuses whatever it receives with its local feature and produces the final 3D detection $\hat{Y}_t$, which is scored against the ground truth Y_t.

We adopt a receiver-driven architecture: the ego evaluates its own perception and channel state, decides which granularity it needs, and signals the choice with a 2-bit request. We treat the request as an application-layer extension associated with the ETSI Cooperative Awareness Message [21] and Collective Perception Message [22] services, carried over IEEE 802.11bd or NR sidelink [6]; the standards integration itself is outside the scope of this paper. The request is a fixed control overhead, identical for all three actions, and is therefore not charged to any action's payload. It is sent on every frame, including when the ego selects E; what E removes is the cooperative perception payload, not the request.

B. Message Candidates

The ego selects one action per frame,

$$s_t \in \mathcal{A} = \{E, L, F\}, \tag{1}$$

where E is ego-only operation—the collaborator transmits no cooperative perception payload and the ego detects from its own LiDAR—L is the collaborator's detected boxes, and F is the collaborator's quantised bottleneck feature tensor.

The candidate action set is $\mathcal{A} = \{E, L, F\}$. The frozen policy selected during the development protocol has support $\{E, L\}$; feature-level transmission remains in the candidate set and in the unified evaluation framework.

F's payload is $1,244\times$ that of L on the development split, so Fixed F is infeasible as an always-on policy, sitting at $5.0\times$ the primary budget. That arithmetic does not by itself exclude requesting F occasionally, because the budget constrains the mean payload rather than each frame; what leaves it out here is that the frozen policy does not select it (Section VII-A). F is measured and charged through the identical chain throughout, as an over-budget fixed-action reference and as the object of the transport study of Section VI-E.

E is a first-class action, not merely a failure mode. It costs no perception payload and cannot fail in transit, which is right both where the ego's own detection already suffices and where the channel is too poor for a message to be worth its cost.

The object-level message is a compact set of local detections, $m_t^L = \{(b_k, c_k, p_k)\}_{k=1}^{N_t}$, with b_k the 3D box, c_k the class and p_k the confidence. The feature-level message is the encoder's bottleneck tensor, $739,200$ elements, quantised to $w = 8$ bits per element before transmission. No modulation-order variant is part of the action space: the selector chooses what to send, whereas the constellation is a transport parameter of the link.

C. Channel Model and Transport

Let γ_t be the estimated channel quality at frame t in dB, available from pilot-based estimation or link-layer reports as already produced by IEEE 802.11bd and 5G NR sidelink stacks [6]. Messages are carried by a rate-1/2 LDPC code with information block $K = 500$ and codeword length $n = 1000$, modulated with 16-QAM.

A message is fragmented into packets of $8,000$ payload bits with a 320-bit header following common V2X framing assumptions [6], [7], and each packet is coded into LDPC codewords. Each codeword succeeds or fails independently at the frame's operating point with per-codeword loss probability p. Three delivery regimes are evaluated:

- fragment-aware partial recovery (the primary regime for the feature message): the receiver reconstructs the tensor from the codewords that arrive and zero-fills the elements carried by codewords that do not;
- packet loss: the same rule applied at packet granularity, a post-freeze transport sensitivity;
- message all-or-nothing: the message is usable only if all of its codewords decode, a pessimistic sensitivity arm for the feature message.

The delivery semantics of the two cooperative actions are not the same, and the distinction is load-bearing:

> Late fusion uses message-level delivery. For frame t, all $N_{\text{cw},L,t}$ codewords must be received for the collaborator box message to be available, giving $q_{L,t} = (1-p)^{N_{\text{cw},L,t}}$; otherwise execution falls back to the ego-only result. In contrast, the primary feature-fusion regime performs codeword-level partial recovery and does not fall back to ego-only when only part of the feature message is lost.

D. Effective Task Utility under Channel Corruption

Let F_t^s denote the realised frame F_1 when action s is executed at frame t.[1] For the object-level action the message-level rule above gives

$$\begin{aligned} F_t^L(p) &= q_{L,t}\, F_t^{L,\text{clean}} + \big(1 - q_{L,t}\big) F_t^{\text{ego}}, \\ q_{L,t} &= (1-p)^{N_{\text{cw},L,t}}, \end{aligned} \tag{2}$$

[1] Here, F_t^s denotes the frame-level F_1 utility under action $s \in \{E, L, F\}$. Thus, F_t^F is the frame-level F_1 obtained when the feature-level action F is executed.

where F_t^{ego} is the ego-only frame F_1. For the feature-level action under partial recovery there is no such mixture, because a partly damaged tensor is still fused: $F_t^F(p)$ is measured on the reconstructed tensor at each sampled p and interpolated piecewise-linearly in raw p between the sampled nodes, with the endpoints locked to the clean measurement at $p = 0$ and the all-lost measurement at $p = 1$. No monotone correction is applied, because some masks raise a frame's F_1 by removing false positives and smoothing that away would erase a measured effect (Section VI-E).

Payload accounting is separate from this expectation: communication is charged for the message that is attempted and is never multiplied by the success probability. A failed delivery does not refund symbols already spent on the channel.

E. Communication Cost

Payload is a per-frame quantity, not a constant per action:

$$B_{t,s} = \begin{cases} 0, & s = E \text{ (or no collaborator)}, \\ B_{L,t}, & s = L, \\ B_F, & s = F, \end{cases} \tag{3}$$

in channel-use-equivalent mega-symbols (Msym). $B_E = 0$ is the cooperative perception payload, not the total channel use of the frame: the request itself is a fixed control overhead that every action pays identically (Section III), so it cancels from absolute payload differences; the reported relative savings refer to cooperative-perception payload only. It is excluded from $B_{t,s}$ throughout. The feature payload follows from the codeword count rather than from $(\text{info} + \text{header})/R/\log_2 M$, because the direct route ignores codeword padding: $739,200$ elements at 8 bits give $5,913,600$ information bits, which packetise into 740 packets carrying $236,800$ header bits, hence $N_{\text{cw}} = 12,567$ codewords and $B_F = 3.14175$ Msym per frame. The direct route would have given 3.0752 Msym, a difference that is pure padding. The object-level payload is charged through the identical chain from each frame's own box count, at 184 bits per box, ranging from 0.00050 to 0.00525 Msym, i.e. 2 to 21 codewords. On the development split, the collaborator transmits an average of 24.08 boxes per frame, producing 10.10 LDPC codewords and 0.00253 Msym. The post-fusion late-detection output contains 28.40 boxes on average, but this value is not used for payload accounting.

The mean cost over T frames is $\bar{B} = (1/T) \sum_t B_{t,s_t}$.

F. Task–Communication Objective

Let z_t be the ego-side cue vector and $(\hat{\gamma}_t, c_t)$ the estimated channel state. The selector is a map

$$s_t = g(z_t, \hat{\gamma}_t, c_t). \tag{4}$$

G. Derivation: Constrained Optimisation and Rate–Distortion View

The design problem is

$$\max_g \ \mathbb{E}[F_t^{s_t}] \quad \text{s.t.} \quad \mathbb{E}[B_{t,s_t}] \le \beta B_F, \tag{5}$$

where the budget is expressed normalised to the feature payload. We retain the three budget tiers $\beta \in \{0.10, 0.20, 0.30\}$, i.e. 0.31418, 0.62835 and 0.94252 Msym per frame; the primary cell is $\beta = 0.20$. The budget is a mean constraint across frames, not a per-frame cap—under a per-frame cap any $\beta < 1$ would make F unselectable on every frame and the problem would degenerate.

Introducing a multiplier $\lambda \ge 0$ gives the pointwise relaxed optimum

$$s_t^\star(\lambda) = \arg\max_{s \in \mathcal{A}} \big(F_t^s - \lambda B_{t,s}\big), \tag{6}$$

with ties broken $E \succ L \succ F$. The learned selector is trained against Eq. (6), and the constrained problem is addressed by sweeping λ over the family.

In rate–distortion terms, Eq. (2) and the measured $F_t^F(p)$ express a channel-induced distortion that grows with the loss rate, while payload is the rate. The selector therefore trades rate against rate-conditional distortion as a function of channel state—joint source–channel structure expressed at the granularity-action level rather than at the symbol level. This yields the design principle the paper rests on: the most informative message is not always the most useful message under communication constraints.

IV. Proposed Method

A. Overview

Instead of transmitting a fixed representation on every frame, the ego constructs an ego-local cue vector, reads its channel estimate, runs a lightweight selector, signals the chosen granularity with the 2-bit request, and the collaborator complies. Fig. 1 shows the data flow as implemented, including which quantities exist before the request is issued.

B. Ego-Side Online Task Cues

The cue vector is $z_t \in \mathbb{R}^{23}$ under the frozen schema v2_ego_local_23d: 21 ego-local task and availability cues—19 statistics of the ego's own LiDAR sweep (point counts, range statistics, near/mid/far and per-sector occupancies, and three range densities), the ego's own detected box count, and a binary collaborator-availability flag—together with 2 channel cues. The complete field list with a code location per row is in the supplementary material.

Every cue is computed from the ego vehicle's own observation and its own detection stream, and every field is computable before the request is issued. This is a direct consequence of the receiver-driven architecture: nothing in z_t depends on a message the ego has not yet asked for. Ground truth, task metrics, oracle labels and delivery outcomes are excluded from the schema by construction, and the cue schema is verified against its code-defined sources.

C. Channel Quality Cue

The channel state is the estimated SNR $\hat{\gamma}_t \in \mathbb{R}$ in dB and a binary channel-type indicator c_t. In a deployed V2X stack $\hat{\gamma}_t$ comes from pilot-symbol estimation already produced by 802.11bd or NR sidelink receivers [6] and c_t can be inferred from link-layer multipath indicators. Both are treated as estimates, and the results below are conditional on their being available at the ego.

The role of $(\hat{\gamma}_t, c_t)$ differs from that of the task cues. Task cues estimate whether the frame needs richer information; channel cues estimate whether such information can be carried. A difficult frame does not by itself justify a feature request.

D. Message Branches

The object-level branch transmits the collaborator's detected boxes and is charged per frame from its own box count. The feature-level branch transmits the encoder's bottleneck tensor quantised to 8 bits per element. Quantisation is part of the transmitted form and its cost is measured: on a 220-frame audit subsample of the development split the mean frame F_1 moves from 0.88808 (float) to 0.88828 (int8), a difference of +0.00020. This is a quantisation audit on that subsample and is not a clean-channel result for the full split.

E. Channel-Aware Semantic Granularity Selector

The policy $g(\cdot)$ is instantiated as a random forest [23] with 400 trees and a minimum of 2 samples per leaf, operating purely on tabular cues. It adds no learnable parameters to the perception backbone and requires no retraining of the detector, which is why a tabular learner is a natural fit; the forest is one realisation of the selection layer, not the contribution.

The selector is trained to imitate the utility-maximising action of Eq. (6), which uses realised task utility and is therefore not deployable; it supplies training labels only.

Feasibility: Mainline feasibility is structural. An action is feasible iff a collaborator is available and a transport response is defined; on a frame with no collaborator the executed action is forced to E. We do not apply a message-level BLER feasibility mask because the quantity it thresholds is the per-codeword erasure probability rather than the probability that the feature message is unusable, and because channel impairment is already reflected in the partial-recovery effectiveness. The feature output is defined and measured at every loss rate, including $p = 1$. No $F_t^F - F_t^{\text{ego}}$ safety threshold is added either: with $\lambda \geq 0$ and $B_F > 0$, $F_t^F \leq F_t^{\text{ego}}$ already implies $U_F \leq U_E$, so the objective handles it and an extra threshold would only add a researcher degree of freedom.

F. Receiver-Side Fusion and Detection

The ego fuses the received message with its local feature, $\hat{Y}_t = D\big(\Psi(F_{\text{ego},t}, m_t^{s_t})\big)$, where Ψ is the fusion module and D the detection head. If $s_t = L$ the receiver performs object-level fusion; if $s_t = F$ it dequantises the received tensor—zero-filling the elements whose codewords were lost—and performs intermediate fusion.

A zeroed collaborator tensor is not the same input as no collaborator at all. The fusion module attends over both branches, so a zero tensor participates and dilutes the ego features. The difference is measured in Section VI-E.

V. Experimental Protocol

A. Dataset and Implementation

We evaluate on the OPV2V dataset [1] using the OpenCOOD codebase, with a PointPillars BEV backbone [20]. One checkpoint serves all three actions, and the field of view and ground truth are identical across them, so an L-versus-F comparison is a comparison of granularity rather than of two separately trained pipelines. Exactly one collaborator is used, selected by a fixed rule, so the payload of a cooperative action is well defined.

In the evaluation, collaborator availability is determined before action selection from the frame's vehicle identities, relative poses and the fixed communication-range rule. In deployment, equivalent neighbour identity and position information can be obtained from periodic cooperative-awareness broadcasts; determining availability therefore does not require receiving the requested perception payload. Table I states the three splits and their roles.

Evaluation is deterministic: the per-sample point shuffle is seeded on (split, scene, frame, cav), and two independent processes reproduce a run bit for bit, verified by a reconstruction-consistency bridge.

B. Message Construction and Payload Accounting

Payload is measured, not declared, and the derivation is given in full in Section III-E and expanded bit by bit in the supplementary material. Two properties matter for reading the results. First, B_F is a constant 3.14175 Msym while $B_{L,t}$ varies per frame with the collaborator's box count, so the payload axis is not a two-point scale. Second, payload is charged for the attempt (Section III-D), which is what makes an undelivered feature message expensive rather than free.

C. Channel Settings

The estimated SNR is swept over an 11-point grid and crossed with two channel types, giving the full deterministic grid of frames × SNR × channel type reported in Table I. Transport is characterised at 8 per-codeword loss rates with 4 independent channel realisations each, and the message regime is evaluated by 200 Monte-Carlo replays.

Across the 11 evaluated SNR points, p_{cw} ranges from 1 to 0 under AWGN and from 0.9835 to 0.0400 under Rayleigh. The AWGN waterfall occurs between 6 and 8 dB, where the measured value falls from 0.74650 to 0.00013; the complete grid is reported in the supplementary material.

TABLE I
Experimental protocol. The development split is where the candidate, the penalty λ and the comparator threshold were chosen; both held-out splits were evaluated only after model and comparator freezing.

split	frames	scenes	grid rows	no-collab. rows	role
validate (dev.)	1,980	9	43,560	0	development: candidate, λ and τ chosen here
Test	2,170	16	47,740	2,618	held out; evaluated only after freezing
Culver-City	550	4	12,100	1,584	held out; evaluated only after freezing (domain shift)

D. Selector Training and Freezing

Model selection is done on the development split only, in this order.

1) Scene-level 9-fold leave-one-scene-out (LOSO) on validate. Folds are scene-disjoint, because frames within a scene are consecutive samples of the same traffic and are far from independent.
2) Candidate and penalty selection. A candidate walk over selector configurations and penalties λ is run under the budget constraint, scored by LOSO.
3) Refit and freeze. The selected configuration—candidate 67 at $\lambda = 0.2$—is refitted on the full development split and frozen, together with the comparator: an SNR threshold at $\tau = 16.5$ dB, the best budget-feasible threshold at the primary cell. Per-tier thresholds are 18.5, 16.5 and 14.5 dB for $\beta = 0.10$, 0.20 and 0.30.
4) One-shot held-out evaluation. Test and Culver-City are evaluated only after model and comparator freezing.

LOSO is the training-time selection procedure and the scene-level bootstrap of Section V-F is the evaluation-time interval procedure; they operate at different stages and neither replaces the other.

E. Baselines

Fixed E, Fixed L and Fixed F bound the action space. The internal decision rules are the SNR-threshold rule and two- and three-scalar hand rules over the same cues, fitted through the identical LOSO walk and budget constraint. The budget-blind oracle selects the utility-maximising action per row and is an upper-bound reference, not a deployable policy. The frozen $\tau = 16.5$ is the primary comparator because it is the only arm sharing CA-TOSG's complete payload definition.

Where2comm is included as an external feature-content baseline [3]. It is evaluated under the same data splits, field of view, ground truth and detection metrics. We report its native communication rate because the evaluated implementation transmits floating-point selected features and does not execute the locked int8 quantisation path. Consequently, it is excluded from bit-level budget-matched claims. No adaptive-policy baseline met the reproducibility bar for inclusion as a policy comparator.

F. Evaluation Metrics

AP@0.5 and AP@0.7 characterise the fixed perception arms, transport responses and the external Where2comm reference. The confirmatory comparison between the frozen selector and its budget-feasible comparator uses scene-equal mean per-frame F_1, with scene-level bootstrap intervals. Communication cost is the mean per-frame channel use defined in Section III-E.

The non-inferiority margin is $\delta = 0.005$ in absolute mean per-frame F_1. It is a preregistered engineering tolerance—how much accuracy may be given up in exchange for a substantive communication reduction—fixed on 2026-08-09, before any evaluation on either held-out split. It is not a safety-certified threshold, and no statistical or regulatory standard is claimed for its value. The confirmatory unit is the scene, not the frame. The headline statistic is therefore the scene-equal mean F_1, and intervals are scene-level bootstraps with 16 Test scenes and 4 Culver-City scenes, covering 47,740 and 12,100 evaluated rows. Every payload figure is reported under two accountings—over all frames, and over frames where a collaborator is available—and the two are never mixed.

VI. Results

A. Fixed-Action Performance and Communication Cost

Table II gives the complete arm set. On the development split the three fixed actions separate as expected in accuracy—ego-only at scene-equal F_1 0.82567, object-level at 0.85854, feature-level at 0.86506—but not at all in cost: Fixed L spends 0.00253 Msym per frame against 3.14175 Msym for Fixed F. In detection terms over the same split, ego-only reaches AP@0.5 0.72055 and object-level 0.88851, while the int8 feature message under clean delivery reaches 0.86994.

Feature-level fusion exceeds object-level fusion by +0.00652 in scene-equal F_1 on that split and costs $1,244\times$ the channel use.

The ordering is metric-dependent. Feature-level fusion exceeds object-level fusion by +0.00652 in scene-equal mean per-frame F_1, but is lower by −0.01857 in development-split AP@0.5. The former is the frozen selector utility, whereas AP globally ranks confidence-scored detections across the split. Therefore, under this checkpoint, F is not uniformly superior to L; its measured advantage is specific to the per-frame F_1 objective used for granularity selection.

B. Primary Held-Out Comparison

The preregistered comparator belongs to the original L/F-capable formulation and is retained unchanged for

TABLE II
All evaluated arms. Development-split rows are where every choice was made. ρ_E, ρ_L and ρ_F denote requested-action shares. On collaborator-unavailable frames, execution is structurally forced to E regardless of the request; the effective utility and realised payload already incorporate this rule. All ratios reported in the text are computed from the underlying unrounded values; table entries are rounded for display.

arm	scene-eq. F_1	payload (Msym)	ρ_E	ρ_L	ρ_F	feasible at $\beta = 0.20$
development split (validate) — where every choice was made						
Fixed E (ego-only)	0.82567	0.00000	1.000	0.000	0.000	✓
Fixed L (object-level)	0.85854	0.00253	0.000	1.000	0.000	✓
Fixed F (feature-level)	0.86506	3.14175	0.000	0.000	1.000	×
CA-TOSG (frozen RF)	0.86050	0.00131	0.457	0.543	0.000	✓
budget-blind oracle	0.88321	1.76778	0.191	0.246	0.562	×
SNR threshold $\tau = 16.5$	0.86289	0.57329	0.000	0.818	0.182	✓
best hand rule (two_scalar_snr8.0_nbox25)	0.87437	1.24792	0.000	0.603	0.397	×
held-out splits — evaluated after freezing						
Test: Fixed E	0.85666	0.00000	1.000	0.000	0.000	✓
Test: Fixed L	0.87715	0.00136	0.000	1.000	0.000	✓
Test: Fixed F	0.88085	2.96946	0.000	0.000	1.000	×
Test: CA-TOSG	0.87679	0.00081	0.403	0.597	0.000	✓
Test: $\tau = 16.5$	0.88178	0.54102	0.000	0.818	0.182	✓
Culver-City: Fixed E	0.84642	0.00000	1.000	0.000	0.000	✓
Culver-City: Fixed L	0.87485	0.00251	0.000	1.000	0.000	✓
Culver-City: Fixed F	0.89430	2.73047	0.000	0.000	1.000	×
Culver-City: CA-TOSG	0.87352	0.00147	0.407	0.593	0.000	✓
Culver-City: $\tau = 16.5$	0.87813	0.49850	0.000	0.818	0.182	✓

protocol integrity; the matched-payload E/L comparisons in Section VI-C address the deployed policy's direct comparator question.

Table III and Fig. 2 give the primary cell, $\beta = 0.20$. Relative to the frozen threshold comparator, on Test the realised payload falls by 99.85% and on Culver-City by 99.70%; on both splits the scene-level bootstrap lower bound on ΔF_1 lies outside the preregistered margin of -0.005, so accuracy non-inferiority is not established.

The payload ratio is 669 and 339, and the shortfall on the bound is 0.0024 (Test) and 0.0004 (Culver-City). The criterion is defined on the bound: on both splits the point estimate lies inside the margin and the bound does not.

The Culver-City interval rests on only 4 scenes and is correspondingly coarse; it reproduces the shape of the Test result on much less scene diversity rather than independently corroborating it. Under the second accounting—frames where a collaborator is available, which is 94.52% of Test rows and 86.91% of Culver-City rows—the relative saving is unchanged, so none of it is an artefact of collaborator unavailability.

As a descriptive secondary reference, CA-TOSG reduces payload by 40.68% on Test and 41.30% on Culver-City relative to Fixed L, with scene-equal F_1 differences of -0.00036 and -0.00133, respectively. These held-out fixed-arm summaries are reported only as descriptive references derived from the already unsealed frozen grids. They are not used to select a new comparator, retest the primary criterion, or modify any frozen component. The confirmatory comparator remains the $\tau = 16.5$ rule frozen before either held-out split was evaluated.

TABLE III
Primary cell at $\beta = 0.20$. Δ is CA-TOSG minus the frozen comparator; the criterion is defined on the bootstrap bound, not on the point estimate.

	Test		Culver-City	
	CA-TOSG	$\tau = 16.5$	CA-TOSG	$\tau = 16.5$
Scene-eq. F_1	0.87679	0.88178	0.87352	0.87813
Payload (Msym)	0.00081	0.54102	0.00147	0.49850
ΔF_1	-0.00499		-0.00461	
ΔF_1 LCB	-0.00738		-0.00541	
Payload saving	99.85%		99.70%	
Scenes	16		4	

C. Comparison at Matched Realised Payload

This is a secondary analysis: it was not preregistered and does not replace the criterion of Section VI-B. It compares selection policies at equal communication cost rather than at a shared nominal ceiling.

Every deployable policy in Table IV is placed on CA-TOSG's own realised payload—0.00081 Msym on Test and 0.00147 Msym on Culver-City. The matching is numerical: the largest residual deviation over all policies and both splits is 2.3×10^{-7} Msym. Each policy is a per-frame rule over $\{E, L\}$: a random selection at the matched rate, a threshold on the estimated SNR alone, a threshold on the ego's own detected box count alone, and CA-TOSG, which uses both. A non-deployable oracle that sees the realised per-frame outcome bounds what selection can buy at that payload.

On Test, CA-TOSG achieves a higher scene-equal F_1 than each evaluated deployable comparator at matched realised payload, with every pairwise 95% bootstrap lower

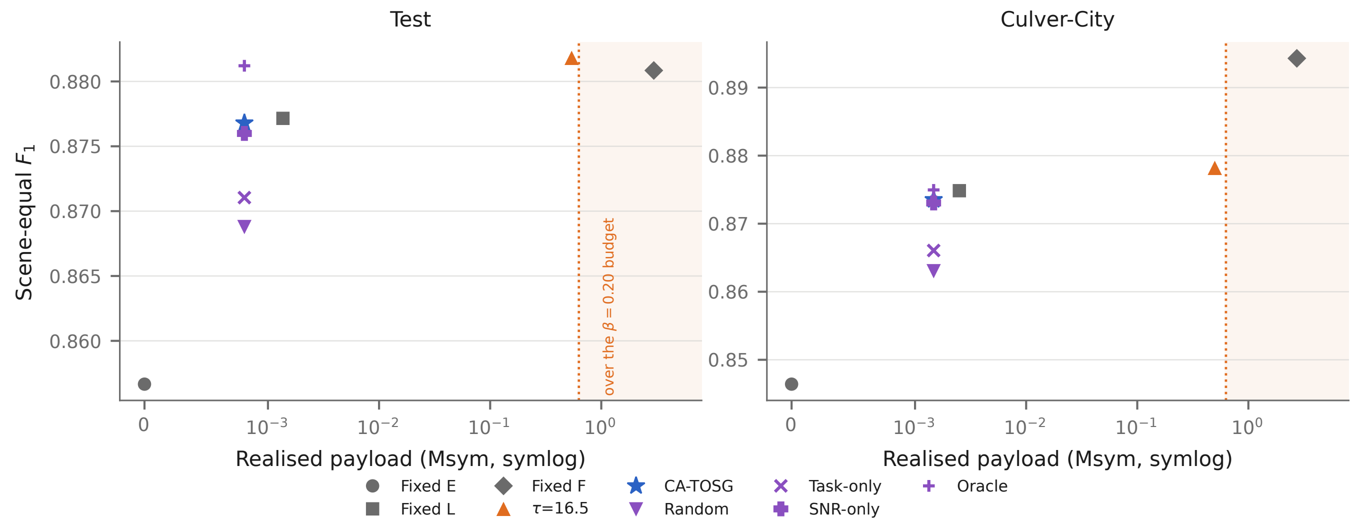

Fig. 2. Scene-equal F_1 against realised payload on each held-out split. Both quantities are shown together, because ranking on either alone misrepresents the trade-off: the arms differ by more than three orders of magnitude in payload—at least 1,090× between Fixed L and Fixed F on either split—and by hundredths in F_1. The dotted line is the $\beta = 0.20$ ceiling and the shaded region is over budget, so Fixed F is visibly not a competitor at the primary budget. The x axis is symlog rather than logarithmic: Fixed E spends exactly zero, which a logarithmic axis cannot show. Random, Task-only, SNR-only and Oracle are the matched-payload policies of Table IV and sit at CA-TOSG's own payload; Oracle is not deployable.

confidence bound above zero. On Culver-City, CA-TOSG has the highest point estimate among the same deployable policies, although the interval against the SNR-only rule includes zero. Culver-City carries 4 scenes, so its intervals are correspondingly coarse. Fig. 3 draws the same differences and their intervals.

Relative to the matched-payload random floor and the non-deployable E/L oracle ceiling, CA-TOSG recovers 64.4% of the attainable selection gain on Test and 87.9% on Culver-City.

Against always-transmitting object-level communication, a second secondary question is how little payload suffices. Among the naturally operating arms that satisfy the primary $\beta = 0.20$ budget and lie within the prespecified $\delta = 0.005$ tolerance of Fixed L, CA-TOSG achieves the lowest realised payload on both held-out splits. Its scene-level bootstrap lower confidence bounds relative to Fixed L are -0.00078 on Test and -0.00252 on Culver-City, both above the -0.005 tolerance boundary. This is reported as a secondary analysis rather than as a replacement for the preregistered comparison.

The comparison set for this second question is the naturally operating arms that satisfy the primary budget: Fixed E, Fixed L, the $\tau = 16.5$ rule and CA-TOSG. Fixed F, the budget-blind oracle and the hand rules exceed that budget; the random, task-only and SNR-only policies are calibrated to CA-TOSG's payload rather than arriving at it.

D. Action Distribution

The frozen selector requests E on 40.3% of Test rows and L on 59.7%, and never requests F ($\rho_F = 0.000$); the Culver-City mix is 40.7% and 59.3%. Section VII-A discusses why the frozen policy does not select it.

Fig. 4 shows where those requests fall. At the low-SNR end the policy requests ego-only almost everywhere; above the AWGN waterfall region of Section V-C, object-level cooperation dominates. Under Rayleigh the transition is gradual and shifts with the ego's own detected box count: the channel state sets where the boundary lies, and the task cues move it.

The reported action proportions are the selector's requested actions. On the 5.48% of Test frames and 13.09% of Culver-City frames without an available collaborator, execution is structurally forced to E regardless of the request; these rows incur zero cooperative payload.

All three budget tiers select the same candidate and the budget never binds. At $\beta = 0.10$, 0.20 and 0.30 the frozen policy spends 0.00131, 0.00131 and 0.00131 Msym for scene-equal F_1 0.86050, 0.86050 and 0.86050—three identical operating points, because on the development split the tightest ceiling 0.31418 Msym is still 240× the 0.00131 Msym the policy actually spends there. There are therefore not three distinct budget points in this study, and this is reported as a limitation of the design rather than as a robustness result.

E. Transport Mechanism

Fig. 5 and the full sweep in the supplementary material characterise the three delivery regimes.

Partial recovery is necessary, not a refinement (development split). Under all-or-nothing message delivery the survival probability of a 12,567-codeword message is $q = 3.463 \times 10^{-6}$ even at the lowest evaluated loss rate $p = 0.001$, i.e. an expectation of 0.006857 surviving frames

TABLE IV
Deployable E/L policies at CA-TOSG's own realised payload, with the non-deployable oracle as a ceiling. ΔF_1 and its bound are CA-TOSG minus that policy, from a scene-level bootstrap. Secondary analysis: not preregistered. Payload is matched numerically to within 2.3×10^{-7} Msym, not exactly.

split	policy	scene-eq. F_1	payload (Msym)	ΔF_1	LCB95	deployable
Test	Random E/L	0.86881	0.00081	+0.00798	+0.00475	✓
	Task-only E/L	0.87104	0.00081	+0.00575	+0.00142	✓
	SNR-only E/L	0.87599	0.00081	+0.00080	+0.00028	✓
	CA-TOSG	0.87679	0.00081	—	—	✓
	Oracle E/L	0.88121	0.00081	-0.00441	-0.00738	—
Culver-City	Random E/L	0.86309	0.00147	+0.01044	+0.00766	✓
	Task-only E/L	0.86606	0.00147	+0.00746	+0.00174	✓
	SNR-only E/L	0.87305	0.00147	+0.00047	-0.00102	✓
	CA-TOSG	0.87352	0.00147	—	—	✓
	Oracle E/L	0.87496	0.00147	-0.00144	-0.00275	—

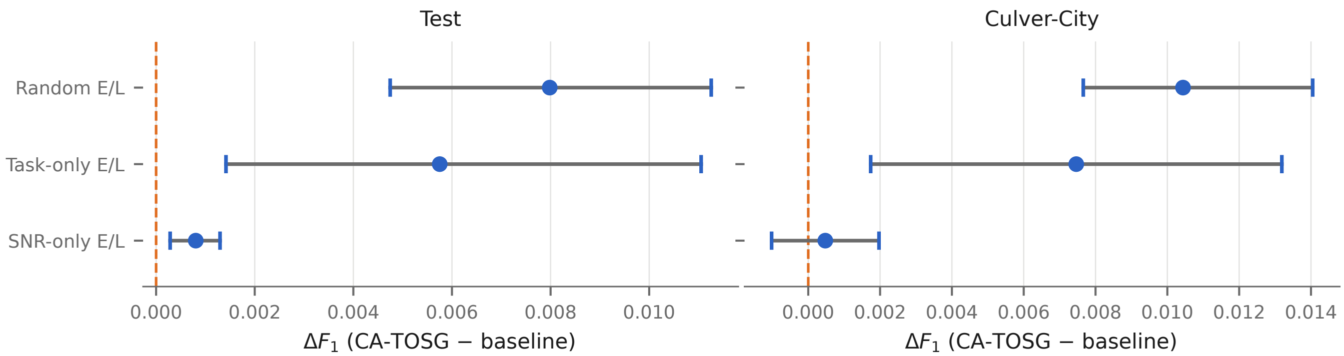

Fig. 3. The same differences as Table IV, drawn: ΔF_1 between CA-TOSG and each deployable baseline at matched realised payload, with 95% scene-level bootstrap intervals and a dashed line at zero. Secondary analysis, not preregistered. Payload is matched numerically to within 2.3×10^{-7} Msym, not exactly. Culver-City carries 4 scenes, so its intervals are coarse—visibly so against the SNR-only rule, whose interval crosses zero. The non-deployable oracle is not shown.

over the whole development split. That regime therefore measures q, not how the feature action performs, and any conclusion drawn from it would be a conclusion about message length.

Where a loss falls matters, at fixed loss amount (development split). Conditioning on exactly equal codeword-loss counts across replicates, $\Delta F_1 \neq 0$ on 28.48% of matched pairs under fragment-aware recovery (1,731 pairs over 1,201 frames; Wilson 95% lower bound 26.40%) and on 25.57% under packet loss (704 pairs, 583 frames, lower bound 22.48%). The sufficiency rule for this test was fixed before the selection ran. We report this at the level it supports—loss position affects task performance—and do not claim the stronger form, that position matters more than amount, because no threshold for that comparison was preregistered.

Zero-tensor fusion is not ego-only inference (development split). At $p = 1$ every codeword is lost and the fused tensor is all zeros, yet the result differs from ego-only detection in both directions: AP@0.5 moves by +0.01195, AP@0.7 by −0.01789 and mean frame F_1 by −0.02658 (from 0.78929 to 0.76271), and the two outputs have identical boxes on 0 of the development split's frames. The signs disagree across metrics, so the two states are not interchangeable in an evaluation that models delivery failure.

F. External Baseline

Fig. 6 places Where2comm on its native axis. At a communication threshold of 0.02 it reaches AP@0.5 0.91708 on Test and 0.83195 on Culver-City (AP@0.7 0.82989 and 0.71588) while retaining 5.2% and 9.1% of features. The complete 24-point sweep is in the supplementary material.

Where2comm uses a separately trained epoch-50 checkpoint and is reported as an end-to-end external system reference rather than as a controlled ablation of semantic granularity. Its held-out AP values must not be numerically ranked against the development-split AP values of the E/L/F arms as though they came from the same split and checkpoint. The confirmatory CA-TOSG comparison is instead defined on scene-equal F_1 against the frozen threshold policy. Where2comm answers a separate question: how much detection performance an external feature-content selection system retains at each native communication rate.

The horizontal axis is the fraction of features retained, not a bit budget, and it is not converted into Msym anywhere in this paper. Doing so would require the

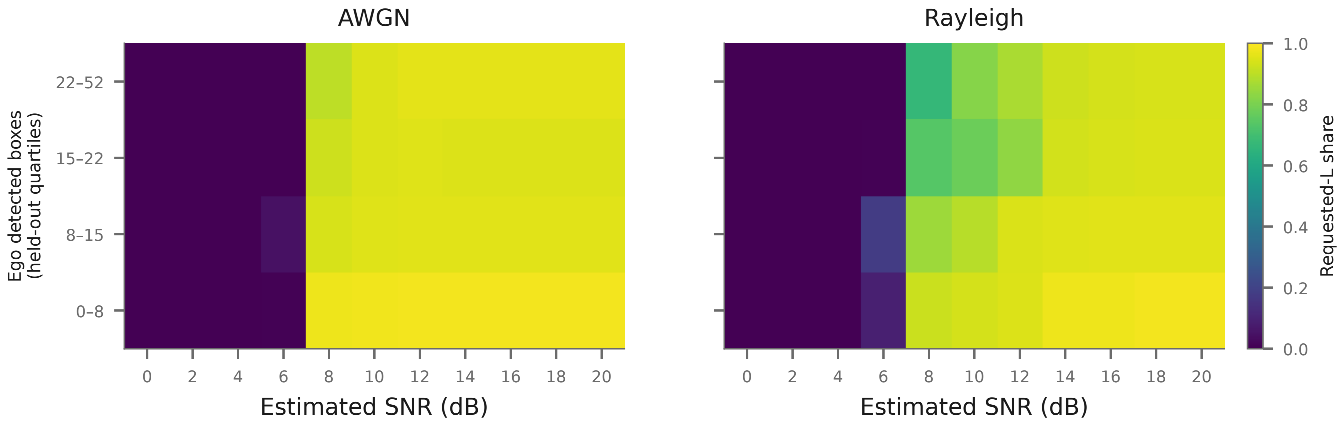

Fig. 4. Where the frozen policy requests object-level cooperation, pooled over both held-out splits. Colour is the requested-action share of L; frames without an available collaborator are structurally executed as E and are counted here by what was requested, never by what was executed. The horizontal axis is the 11 protocol SNR grid points and the vertical axis is the ego detected box count in quartiles of the pooled held-out rows, both fixed before plotting; cells are drawn without smoothing or interpolation, so no structure is shown between points that were evaluated.

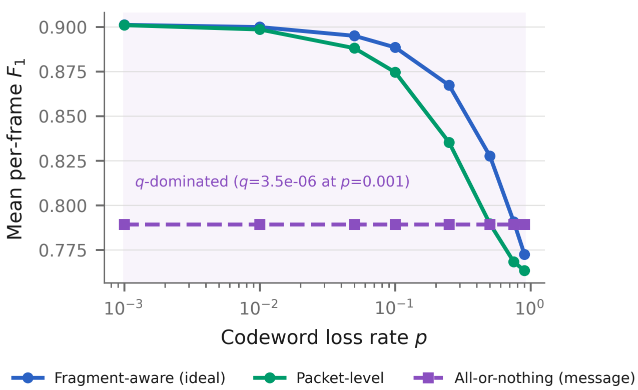

Fig. 5. Delivery regimes on the development split. Fragment-aware partial recovery degrades gracefully; all-or-nothing message delivery is governed by the survival probability q rather than by how the feature action performs.

quantisation, packetisation and coding chain that this implementation does not execute, and the resulting number would describe a system that was never run.

VII. Discussion and Limitations

At matched realised payload the channel state carries most of the decision: the SNR-threshold rule reaches 0.87599 on Test and 0.87305 on Culver-City against CA-TOSG's 0.87679 and 0.87352. The two arms differ in model class as well as in inputs—a single threshold against a random forest—so the difference between them is not attributable to the ego-local cues alone.

All three budget tiers select the same candidate and the budget never binds, so this study does not separate three operating points. Under the same budget ceiling the two feasible policies occupy different points: the threshold rule attains higher scene-equal F_1 and CA-TOSG spends less realised payload.

A. Why the Frozen Policy Operates over E/L

Fixed F costs $1,244\times$ the object-level message (3.14175 Msym against 0.00253 Msym on the development split), which is $5.0\times$ the $\beta = 0.20$ ceiling of 0.62835 Msym. Because the budget constrains the mean payload rather than each frame (Section III-G), that arithmetic excludes F as an always-on policy but not as an occasional request. F is the utility-maximising action on 56.2% of development-split grid rows (24,502 rows), exceeding E and L there by a mean net gain of 0.04890, against which the frozen penalty charges $\lambda B_F = 0.62835$ Msym—a penalty-to-gain ratio of 12.9. The break-even penalty is 0.01556 conditional on the rows where F is best and 0.0048 averaged over the whole grid, each value carrying its conditioning set; both fall between two points of the preregistered penalty grid, so no candidate on that grid selects F. A finer scan of that interval is an exploratory development-split diagnostic reported in the supplementary material; it does not establish how a retrained selector would behave in the break-even region.

B. Other Limitations

Culver-City rests on 4 scenes. The bootstrap resamples them, so its interval is coarse, and Culver-City is a simulated domain shift rather than real-road data.

One ego–collaborator pair. The payload of a cooperative action is defined against exactly one collaborator; a deployment serving several would have to re-derive both the payload and the budget.

An independent codeword-erasure abstraction. Each codeword succeeds or fails independently at the frame's operating point; correlated fading, burst erasures and imperfect channel estimation are not modelled.

The recovery asymmetry between L and F is a modelling choice. The object-level message is delivered at message level, so a single lost codeword costs the whole message,

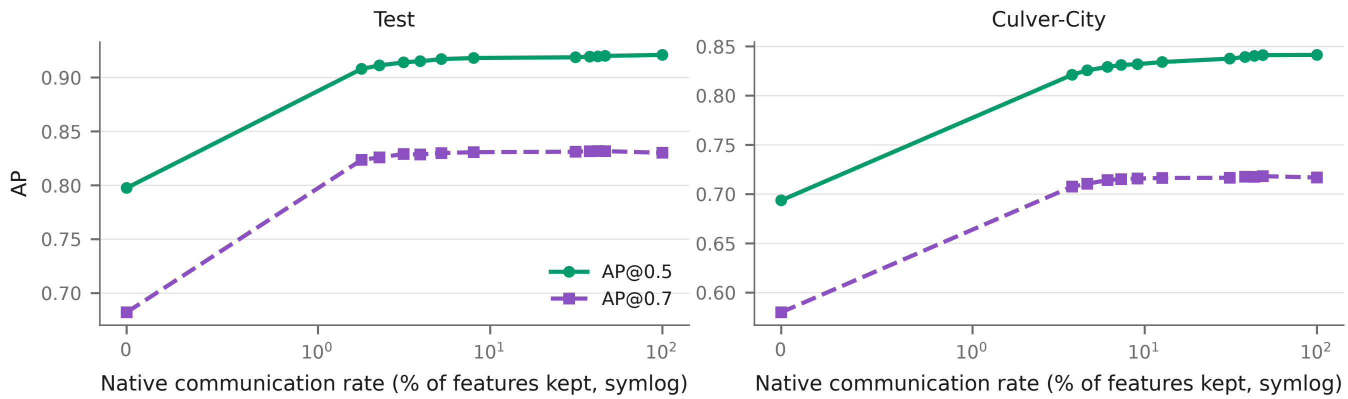

Fig. 6. Where2comm on its native communication-rate axis. The horizontal axis is the fraction of features retained, not a bit budget.

whereas the feature message is recovered codeword by codeword. That asymmetry follows from how each message is framed here, not from a property of object-level data: encoding the boxes as independently decodable groups would make L degrade gracefully too, which would change its effective utility, the oracle labels derived from it, and therefore the selector's decisions. That transport design is not evaluated in this paper.

Selector-only latency. The selector's own inference cost is measured; total end-to-end system latency, including any cost of keeping the cooperative pipeline resident, is not.

One dataset family and one detector. Every result here is conditional on OPV2V and on a single PointPillars checkpoint.

A collaborator-availability mismatch across splits. The development split contains a collaborator on every frame, whereas 5.48% of Test frames and 13.09% of Culver-City frames do not. These rows do not test learned action selection because execution is structurally forced to E and all cooperative actions have zero realised payload. Nevertheless, the difference limits interpretation of the cross-domain action proportions.

VIII. Conclusion

Three transmission granularities were placed under one detector, one field of view and one ground truth, and charged through one measured transport chain. Under the mean per-frame communication budget, the frozen policy operates over ego-only and object-level actions, and CA-TOSG makes that choice per frame from ego-local cues and an estimated channel state; feature-level transmission is retained as an over-budget upper bound.

In a secondary analysis at matched realised payload, CA-TOSG achieves a higher scene-equal F_1 than each evaluated deployable comparator on Test, with every pairwise bound above zero, and the highest point estimate on Culver-City, where the interval against the SNR-only rule includes zero. Against the frozen threshold comparator communication falls by a factor of 669 on Test and 339 on Culver-City, while the preregistered accuracy non-inferiority is not established on either. The conditions are one dataset family, one detector, a single ego–collaborator pair, and an independent codeword-erasure abstraction.

The mechanism results are the part most likely to transfer. Partial recovery is not an optimisation but a precondition for studying feature-level transport at realistic message lengths; loss position changes the task outcome at fixed loss amount, so a transport model that tracks only how much was lost is incomplete; and a zeroed collaborator tensor is not an ego-only input, so the two must be distinguished in any evaluation that models delivery failure. Each of these is a statement about the transport–perception interface rather than about one policy, and each was measured under a protocol in which the detector, the field of view and the ground truth were held fixed.

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